Calculus of Variations Midterm Examination

Reference Solutions | October 13, 2026

The formulas follow the corrected reading of the handwritten source. The proofs and notation follow the supplied tutorial materials: stationary curves are found from the Lagrange equation, and minimality is checked by the second variation, the Legendre condition, and the Jacobi theorem.

Problem 1. Stationary Curve and Global Minimum

1. Lagrange equation

$$L(t,x,v)=e^{2t}(v^2-x^2),\quad L_x=-2e^{2t}x,\quad L_v=2e^{2t}v.$$

$$\frac{d}{dt}L_v-L_x=0 \quad\Longrightarrow\quad \ddot{x}+2\dot{x}+x=0$$

The characteristic equation has the repeated root -1, hence $x(t)=(A+Bt)e^{-t}$. From $x(0)=0$ and $x(1)=e$, we obtain $A=0$ and $B=e^2$. Thus the only admissible stationary curve is

$$x_*(t)=e^2te^{-t}=te^{2-t}.$$

2. Second variation

Following the tutorial notes, for every admissible variation h with $h(0)=h(1)=0$,

$$\delta^2I(x_*)[h]=\int_0^1\left(L_{xx}h^2+2L_{xv}h\dot{h}+L_{vv}\dot{h}^2\right)dt$$

$$=2\int_0^1e^{2t}(\dot{h}^2-h^2)dt.$$

The Legendre coefficient is $B(t)=L_{vv}=2e^{2t}>0$ on $[0,1]$. It remains to apply the Jacobi theorem.

3. Jacobi equation and conjugate points

With $A(t)=L_{xx}-dL_{xv}/dt=-2e^{2t}$ and $B(t)=2e^{2t}$, the Jacobi equation used in the tutorial material is

$$-\frac{d}{dt}\left(B(t)\dot{h}_1\right)+A(t)h_1=0,$$

$$\ddot{h}_1+2\dot{h}_1+h_1=0,\qquad h_1(0)=0,\quad \dot{h}_1(0)=1$$

Its solution is $h_1(t)=te^{-t}$. Since $h_1(t)>0$ for every t in $(0,1]$, there is no point conjugate to 0 in $(0,1]$. Therefore the Jacobi theorem gives $\delta^2I(x*)[h]>0$ for every nonzero admissible h.

Because the functional is quadratic and $x*$ is stationary,

$$I[x_*+h]-I[x_*]=\tfrac{1}{2}\delta^2I(x_*)[h]>0\qquad (h\neq0).$$

Conclusion. The unique global minimizer is $x*(t)=t\ e^{2-t}$.

Problem 2. Isoperimetric Constraint

1. Lagrange multiplier

For the constraint $\int x\,dt=2$, introduce a constant multiplier λ and use the augmented integrand

$$\widetilde{L}(t,x,\dot{x})=t^3\dot{x}^2+\lambda x.$$

$$\frac{d}{dt}(2t^3\dot{x})=\lambda, \qquad x(t)=A+\frac{B}{t}+\frac{C}{t^2}, \qquad \lambda=-2B.$$

2. Determine the constants

$$A+B+C=4, \qquad A+\frac{B}{2}+\frac{C}{4}=1,$$

$$A+B\ln 2+\frac{C}{2}=2$$

The first two equations give $A=-B/3$ and $C=4-2B/3$. Substitution into the integral constraint yields

$$B\left(\ln 2-\frac{2}{3}\right)=0$$

Since $\ln 2\neq 2/3$, $B=0$. Thus $A=0$, $C=4$, and $\lambda=0$.

$$x_*(t)=\frac{4}{t^2}, \qquad \int_1^2 \frac{4}{t^2}\,dt=2$$

3. Global minimality

Let $x=x_*+\eta$, where $\eta(1)=\eta(2)=0$ and $\int\eta\,dt=0$. Since $t^3\dot{x}_*=-8$, the cross term is $-16[\eta]_1^2=0$. Therefore

$$I[x_*+\eta]-I[x_*]=\int_1^2 t^3\dot{\eta}^2\,dt>0 \qquad (\eta\neq 0).$$

$$I[x_*]=64\int_1^2 t^{-3}\,dt=24$$

Conclusion. The unique global minimizer is $x_*(t)=4/t^2$, and the minimum value is 24.

Questions 1 and 2. First Integral and Second Variation

Question 1. L Independent of Time

Let $I[x]=\int L(t, x, \dot{x})\,dt$ and suppose $L_t=0$. Along a smooth stationary curve satisfying the Euler–Lagrange equation,

$$\frac{d}{dt}(L-\dot{x}L_{\dot{x}})=L_t+\dot{x}\left(L_x-\frac{d}{dt}L_{\dot{x}}\right)=0$$

$$L-\dot{x}L_{\dot{x}}=C.$$

This is the Beltrami identity. Equivalently, $H=\dot{x}L_{\dot{x}}-L$ is constant. It reduces the second-order Euler–Lagrange equation to a first-order relation.

Question 2. Second Variation

Let $x_\varepsilon=x+\varepsilon\eta$, with $\eta(a)=\eta(b)=0$ for fixed endpoints. By definition,

$$\delta^2 I[x;\eta]=\frac{d^2}{d\varepsilon^2}I[x+\varepsilon\eta]|_{\varepsilon=0}.$$

Differentiating twice under the integral sign gives

$$\delta^2 I[x;\eta]=\int_a^b (L_{xx}\eta^2+2L_{x\dot{x}}\eta\dot{\eta}+L_{\dot{x}\dot{x}}\dot{\eta}^2)\,dt.$$

If the coefficients are differentiable along x, integration by parts gives

$$\delta^2 I=\int_a^b (P\dot{\eta}^2+Q\eta^2)\,dt,\quad P=L_{\dot{x}\dot{x}},\quad Q=L_{xx}-\frac{d}{dt}L_{x\dot{x}}.$$

$$I[x+\varepsilon\eta]=I[x]+\varepsilon\delta I[x;\eta]+\frac{\varepsilon^2}{2}\delta^2 I[x;\eta]+o(\varepsilon^2).$$

At a minimum, $\delta I=0$ and $\delta^2 I\geq 0$ for every admissible η. The Legendre condition $P\geq 0$ is necessary, but by itself is not sufficient.

Question 3. The Catenary

1. The isoperimetric problem

Following the tutorial derivation, let y=y(x) describe a homogeneous inextensible chain of linear density ρ. Its endpoints are fixed, and the gravitational field is uniform. Apart from the constant factor ρg, minimizing the potential energy means minimizing

$$I(y)=\rho g\int_a^b y\sqrt{1+(y')^2}\,dx,$$

under the prescribed-length constraint

$$J(y)=\int_a^b \sqrt{1+(y')^2}\,dx=J_0$$

By the isoperimetric multiplier theorem, a minimizing curve satisfies the Lagrange equation for the augmented integrand

$$F(y,v)=y\sqrt{1+v^2}+\lambda\sqrt{1+v^2}=(y+\lambda)\sqrt{1+v^2}.$$

2. First integral

The integrand F does not depend explicitly on x. Hence the energy integral is constant:

$$H=y'F_v-F=-\frac{y+\lambda}{\sqrt{1+(y')^2}}=\text{constant}.$$

Squaring this identity gives

$$(y+\lambda)^2=H^2(1+(y')^2).$$

3. The parameter p used in the tutorial

Introduce p by

$$\frac{dy}{dx}=\sinh p,\qquad \sinh p=\frac{e^p-e^{-p}}{2}.$$

Then the first integral becomes

$$(y+\lambda)^2=H^2\cosh^2 p,\qquad y=\pm H\cosh p-\lambda.$$

Differentiating the last relation and also using dy=sinh p dx gives

$$dy=\pm H\sinh p\,dp,\qquad dy=\sinh p\,dx.$$

Therefore dx=±H dp and $x=x_0\pm Hp$. Eliminating p yields

$$y=\pm H\cosh\left(\frac{x-x_0}{H}\right)-\lambda.$$

For a hanging chain take H>0 and the upper sign. Thus

$$y(x)=H\cosh\left(\frac{x-x_0}{H}\right)-\lambda.$$

The constants H, x_0, and λ are determined by y(a)=A, y(b)=B, and J(y)=J_0.

Question 4. Transversality Conditions

Definition and setup

Transversality conditions are the natural endpoint conditions for a variational problem with moving endpoints. We first derive the condition at a moving right endpoint exactly as in the tutorial.

Let $\gamma(\alpha)$ be a one-parameter family of admissible curves $x=x(t,\alpha)$. The left endpoint t_1 is fixed, while the right endpoint moves along a prescribed curve C:

$$t_2 = t_2(\alpha), \qquad x_2 = x_2(\alpha) = x(t_2(\alpha), \alpha).$$

Define

$$F(\alpha) = \int_{t_1}^{t_2(\alpha)} L(t, x(t,\alpha), x'(t,\alpha))\, dt.$$

Differentiate with respect to α

By the Leibniz rule,

$$\frac{dF}{d\alpha} = L(t_2, x_2, x'_2)\frac{dt_2}{d\alpha} + \int_{t_1}^{t_2} (L_x x_\alpha + L_{x'} x'_\alpha)\, dt.$$

Integrating the second term by parts gives

$$\int_{t_1}^{t_2} L_{x'} x'_\alpha\, dt = [L_{x'} x_\alpha]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} L_{x'}\, x_\alpha\, dt.$$

Since the left endpoint is fixed, $x_\alpha(t_1)=0$. At the moving right endpoint, differentiating $x_2(\alpha)=x(t_2(\alpha),\alpha)$ gives

$$\frac{dx_2}{d\alpha} = x'\frac{dt_2}{d\alpha} + x_\alpha(t_2), \qquad x_\alpha(t_2) = \frac{dx_2}{d\alpha} - x'\frac{dt_2}{d\alpha}.$$

Question 4. Transversality Conditions – Continued

Substitution

Substituting the endpoint identity into dF/dα gives

$$\frac{dF}{d\alpha} = \left[(L - L_{x'}x')\frac{dt_2}{d\alpha} + L_{x'}\frac{dx_2}{d\alpha}\right]_{t=t_2} + \int_{t_1}^{t_2}\left(L_x - \frac{d}{dt}L_{x'}\right)x_\alpha\, dt.$$

Introduce the notation used in the tutorial:

$$p = L_{x'}, \qquad H = -L + px'.$$

Then

$$\frac{dF}{d\alpha} = \left[p\frac{dx_2}{d\alpha} - H\frac{dt_2}{d\alpha}\right]_{t=t_2} + \int_{t_1}^{t_2}\left(L_x - \frac{d}{dt}L_{x'}\right)x_\alpha\, dt.$$

Theorem on transversality conditions

If x=x(t) minimizes the functional and its right endpoint moves on C, then x satisfies the Lagrange equation in the interior. Consequently the integral term is zero. Since F′(α)=0 for the minimizing member of the family, the right-end transversality condition is

$$\left[p\frac{dx_2}{d\alpha} - H\frac{dt_2}{d\alpha}\right]_{t=t_2} = 0$$

If the left endpoint also moves along a prescribed curve, the analogous condition is

$$\left[p\frac{dx_1}{d\alpha} - H\frac{dt_1}{d\alpha}\right]_{t=t_1} = 0$$

Thus, at each movable endpoint, the covector (−H, p) annihilates the tangent vector ($dt_i/d\alpha$, $dx_i/d\alpha$) of the prescribed endpoint curve. These are the transversality conditions.